

Data Visualization with Tableau®



Hej, mit navn er **Pradana**.

This is my data science journey!

//dark mode

Excel-PPT guy

Pivot > create table/graph > copy-paste to PPT > present > repeat... for 5 years!

Python introduction

Eye-opener to the data science world

Tableau guy

Tableau introduction!

DA mentor

DA competency mentoring program for new employees in DGTreasury

Healthkathon BPJS 2.0

The first data science competition to bear fruit!

Data-Driven Decision-Making course,
DataViz with Tableau lecture

Tableau instructor

Expanding my expertise in ML,
DL, & LLMs business practice

BI student

**Certified Tableau
BI Analyst**

**Certified Tableau
Desktop Specialist**

2016

2017

2018

2019

2020

2021

2022

2023

2024

2025

2026

Data Visualization

“Data graphics visually display measured quantities by means of the combined use of points, lines, a coordinate system, numbers, symbols, words, shading, and color.

- Tufte, E. R. (2001)

The visual display of quantitative information

“The purpose of visualization is insight, not pictures.

- Shneiderman, B. (2019)

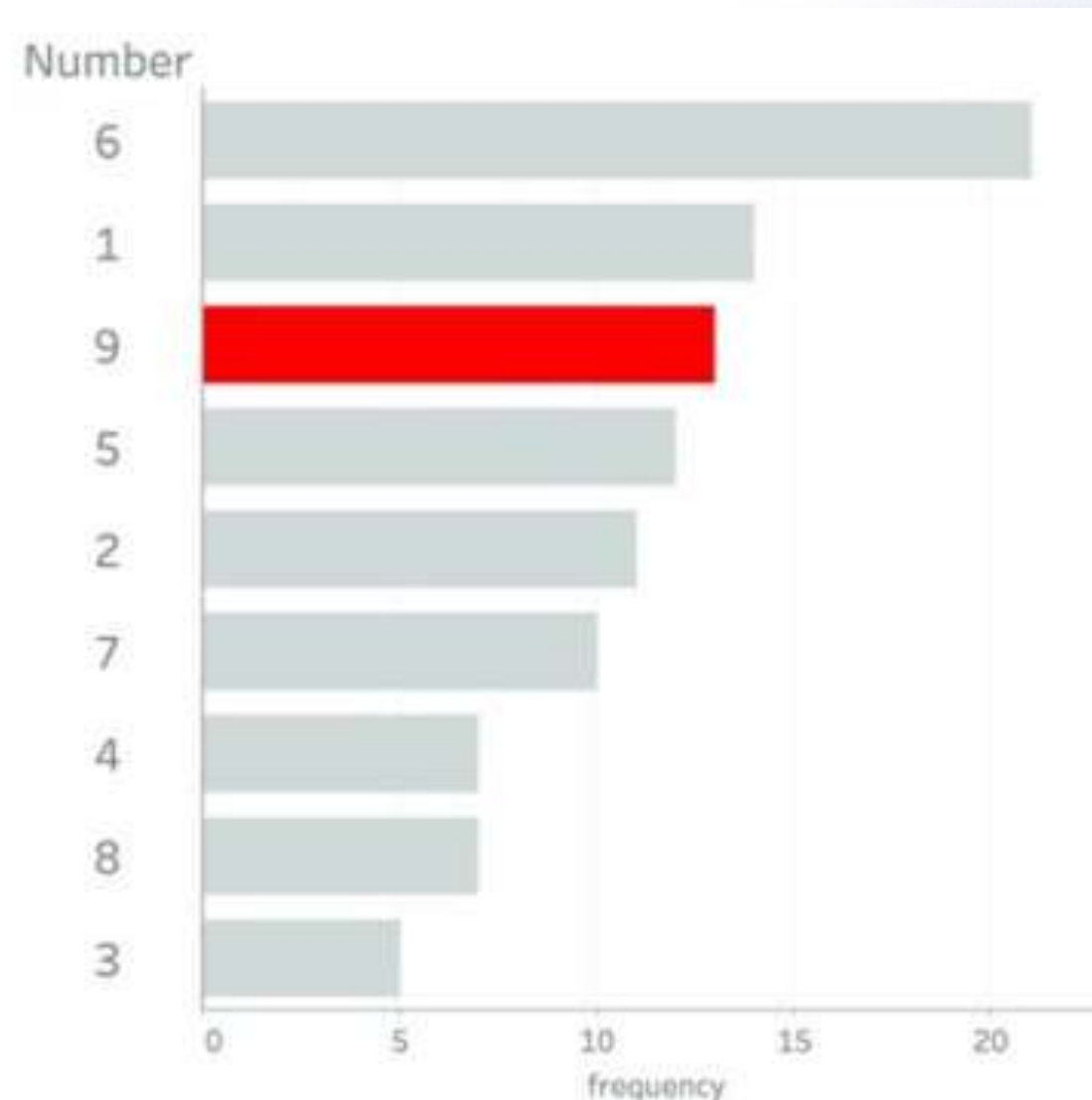
Why data visualization is important?



It helps people **see, interact with, and better understand** data.

The right visualization can bring **everyone on the same page,** regardless of their level of expertise.

How many 9s are here?



2	2	5	6	7	1	1	6	9	1
9	1	7	5	5	5	6	2	5	9
4	5	2	9	6	9	7	6	4	6
8	1	5	7	8	5	6	6	6	7
7	2	3	6	8	9	1	7	9	1
3	8	6	8	4	5	6	9	4	5
4	9	9	2	3	7	1	9	1	2
3	7	8	1	6	1	5	6	1	6
5	6	6	8	6	6	9	1	2	6
3	2	4	2	6	9	4	2	7	1

2	2	5	6	7	1	1	6	9	1
9	1	7	5	5	5	6	2	5	9
4	5	2	9	6	9	7	6	4	6
8	1	5	7	8	5	6	6	6	7
7	2	3	6	8	9	1	7	9	1
3	8	6	8	4	5	6	9	4	5
4	9	9	2	3	7	1	9	1	2
3	7	8	1	6	1	5	6	1	6
5	6	6	8	6	6	9	1	2	6
3	2	4	2	6	9	4	2	7	1

Sekarang hitung jumlah per nomor...

Which month

had the highest sales? Top 3 performing months?

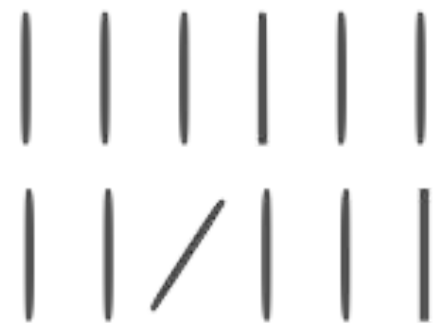
Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
\$44,703	\$20,301	\$58,872	\$36,522	\$44,261	\$52,982	\$45,264	\$63,121	\$87,867	\$77,777	\$118,448	\$83,829

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
\$44,703	\$20,301	\$58,872	\$36,522	\$44,261	\$52,982	\$45,264	\$63,121	\$87,867	\$77,777	\$118,448	\$83,829

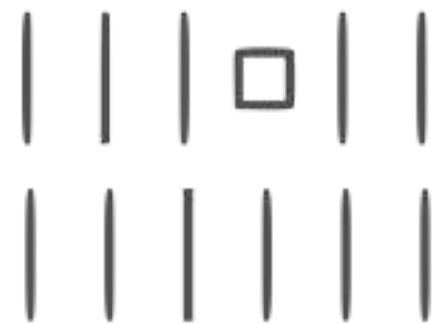
Easier?



Preattentive attributes



Orientation



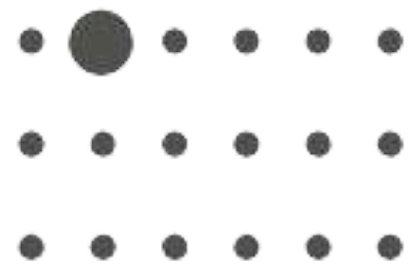
Shape



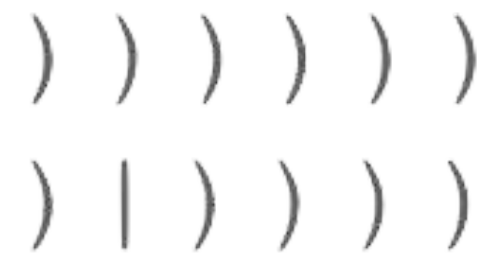
Length



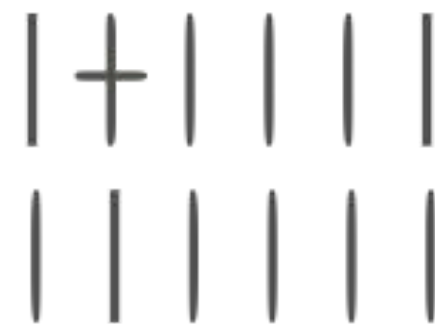
Width



Size



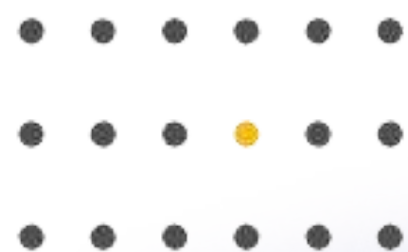
Curvature



Added marks



Enclosure



Hue



Intensity



Position



Motion

Use of color

SEQUENTIAL

color is ordered from low to high



DIVERGING

two sequential colors with a neutral midpoint



CATEGORICAL

contrasting colors for individual comparison



HIGHLIGHT

color used to highlight something

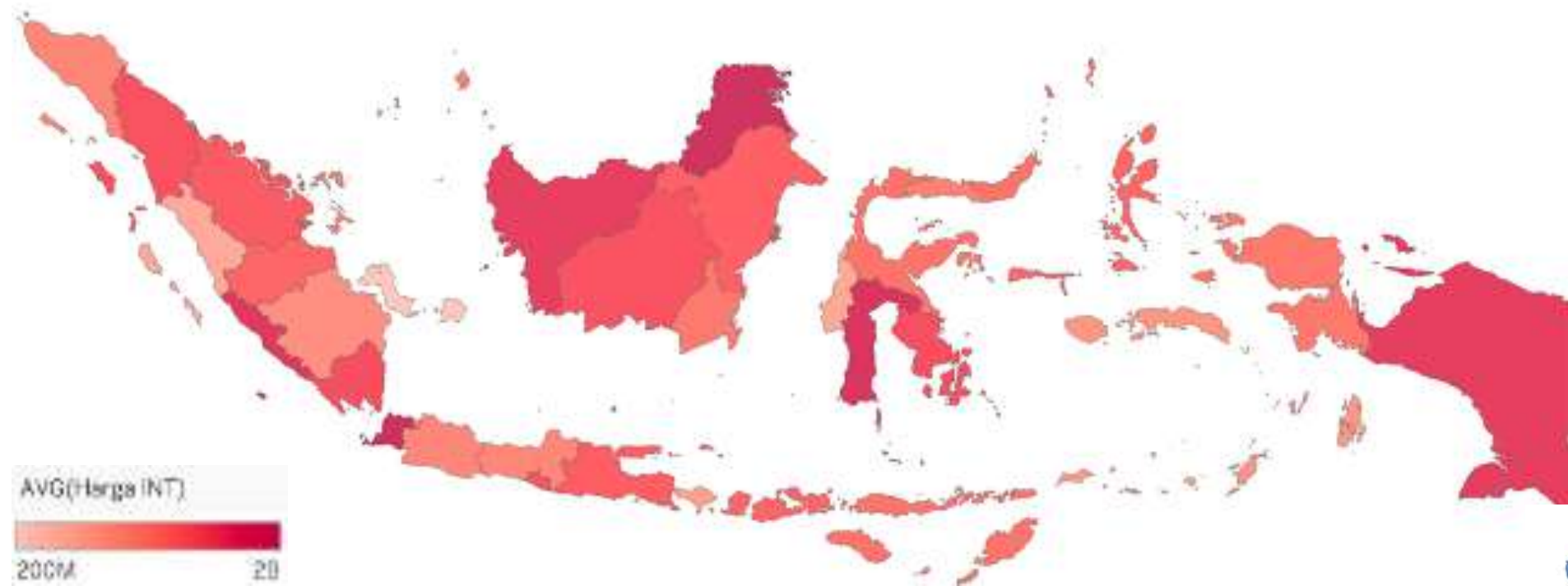


ALERT

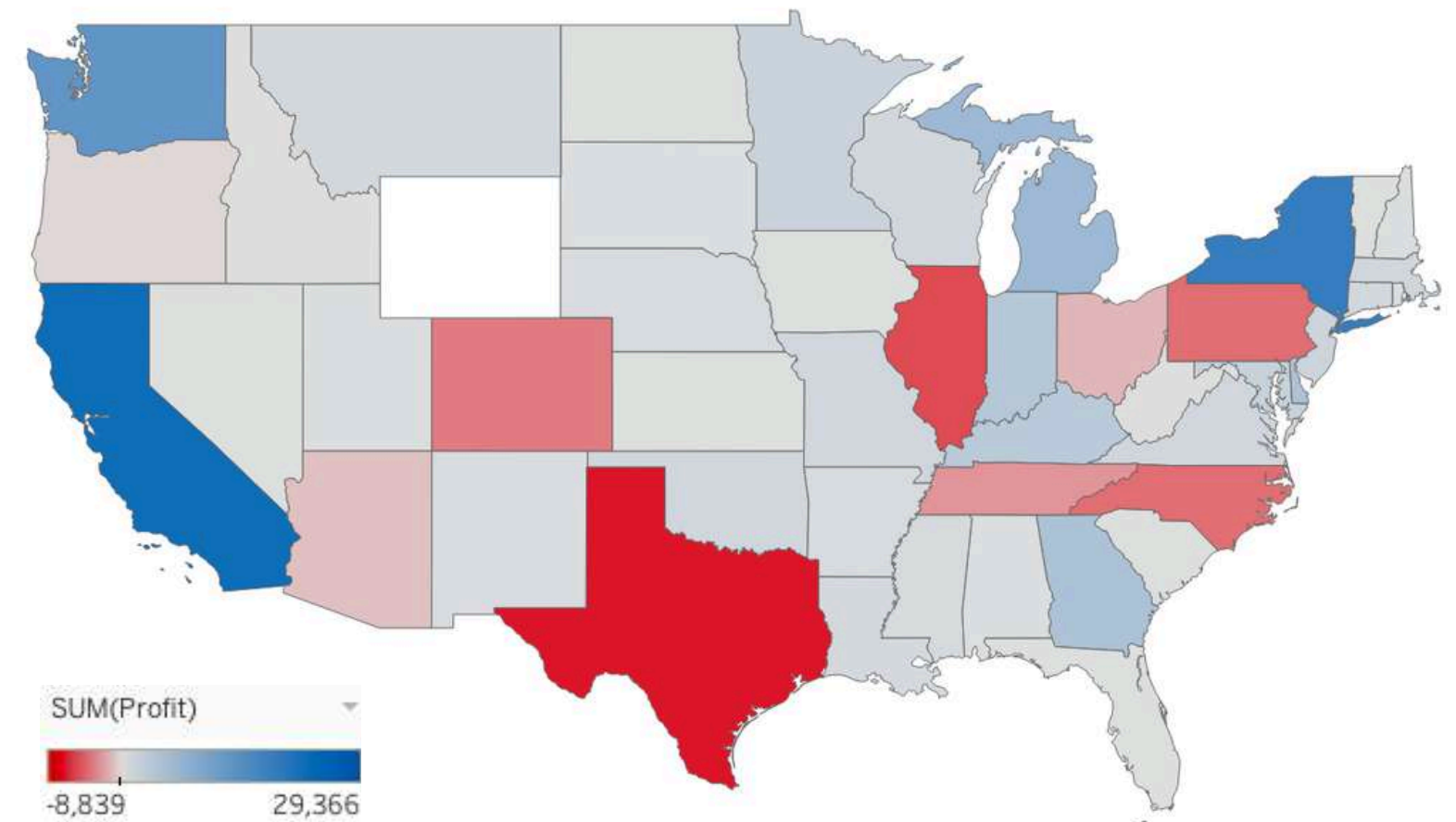
color used to alert or warn reader



Sequential color



Diverging color

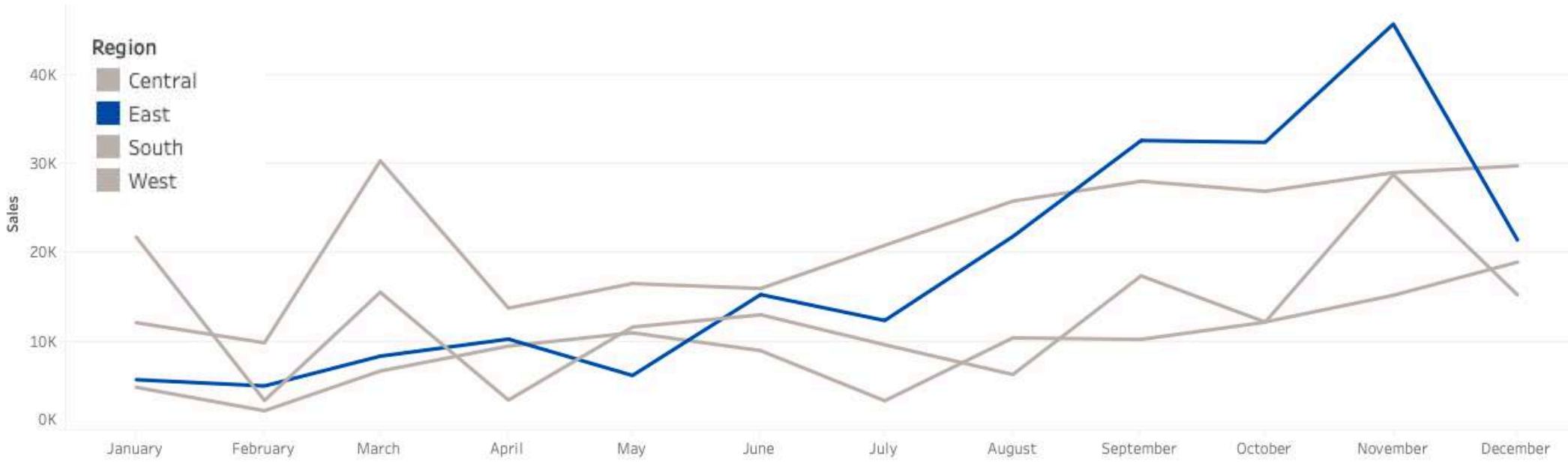


Categorical color

Maintain color identities...
for consistency
for identity
to eliminates confusion



Highlight
color



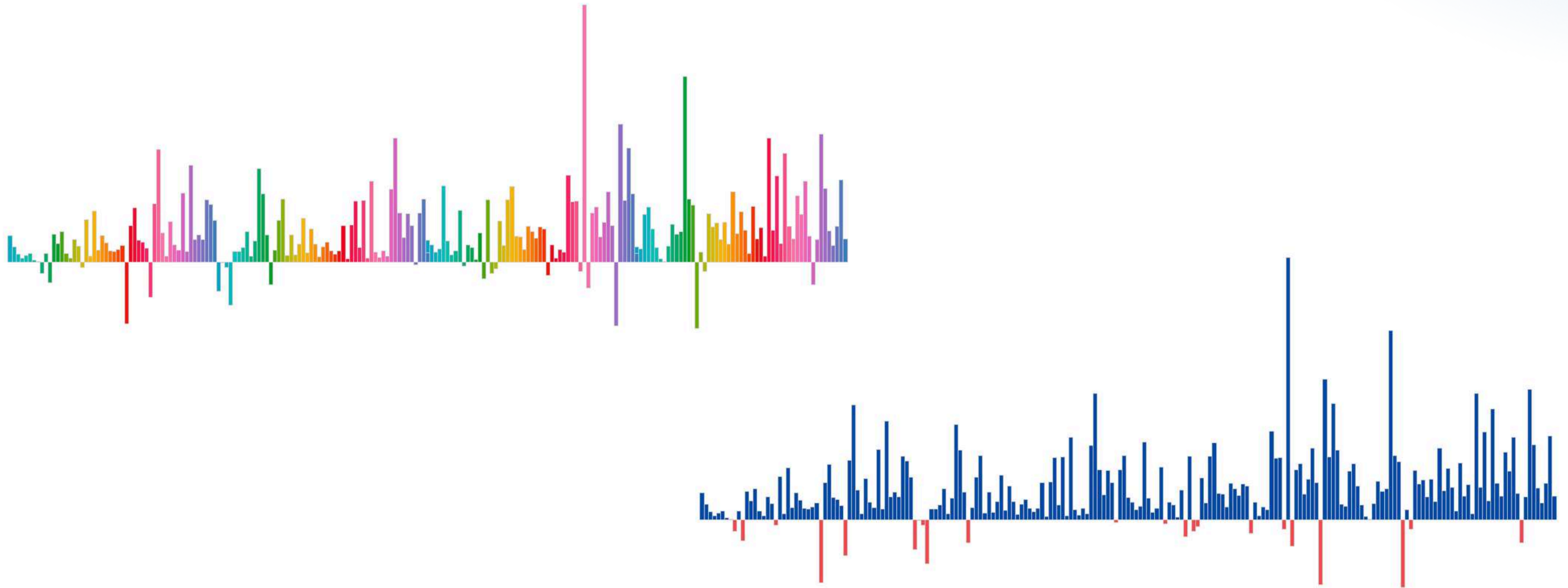
Regional Manager by Profit per Quarter

		Central Roxanne Rodriguez		East Chuck Magee		South Fred Suzuki		West Sadie Pawthorne	
		Profit	Profit Ratio	Profit	Profit Ratio	Profit	Profit Ratio	Profit	Profit Ratio
2022	Q1	139	1.62%	-697	-10.15%	2,674	6.04%	1,979	12.18%
	Q2	969	5.57%	2,050	8.94%	4,446	19.74%	4,220	15.98%
	Q3	-2,004	-4.54%	6,536	18.49%	1,402	8.73%	7,583	15.18%
	Q4	1,435	4.26%	11,121	15.82%	3,358	15.99%	6,473	11.10%
2023	Q1	-220	-1.87%	1,742	9.99%	2,560	15.57%	5,473	22.12%
	Q2	1,535	6.40%	4,368	19.14%	2,289	14.08%	4,008	15.31%
	Q3	2,900	11.84%	5,802	11.38%	4,464	20.80%	3,714	11.08%
	Q4	7,502	17.59%	9,277	14.09%	-995	-5.78%	7,601	13.28%
2024	Q1	-275	-1.36%	3,809	15.16%	5,656	23.63%	2,438	9.54%
	Q2	223	0.86%	8,859	16.77%	3,382	18.40%	4,130	10.36%
	Q3	565	1.64%	3,565	9.21%	3,592	16.60%	8,525	16.79%
	Q4	19,523	28.62%	4,369	6.61%	5,072	17.10%	9,230	12.68%
2025	Q1	3,921	9.68%	2,196	11.59%	3,575	26.20%	14,166	27.16%
	Q2	3,099	11.09%	5,181	16.39%	3,572	12.18%	3,652	7.92%
	Q3	2,744	8.27%	6,292	9.44%	3,243	13.58%	15,267	20.50%
	Q4	-2,192	-4.75%	20,412	20.54%	-1,541	-2.75%	12,339	14.43%

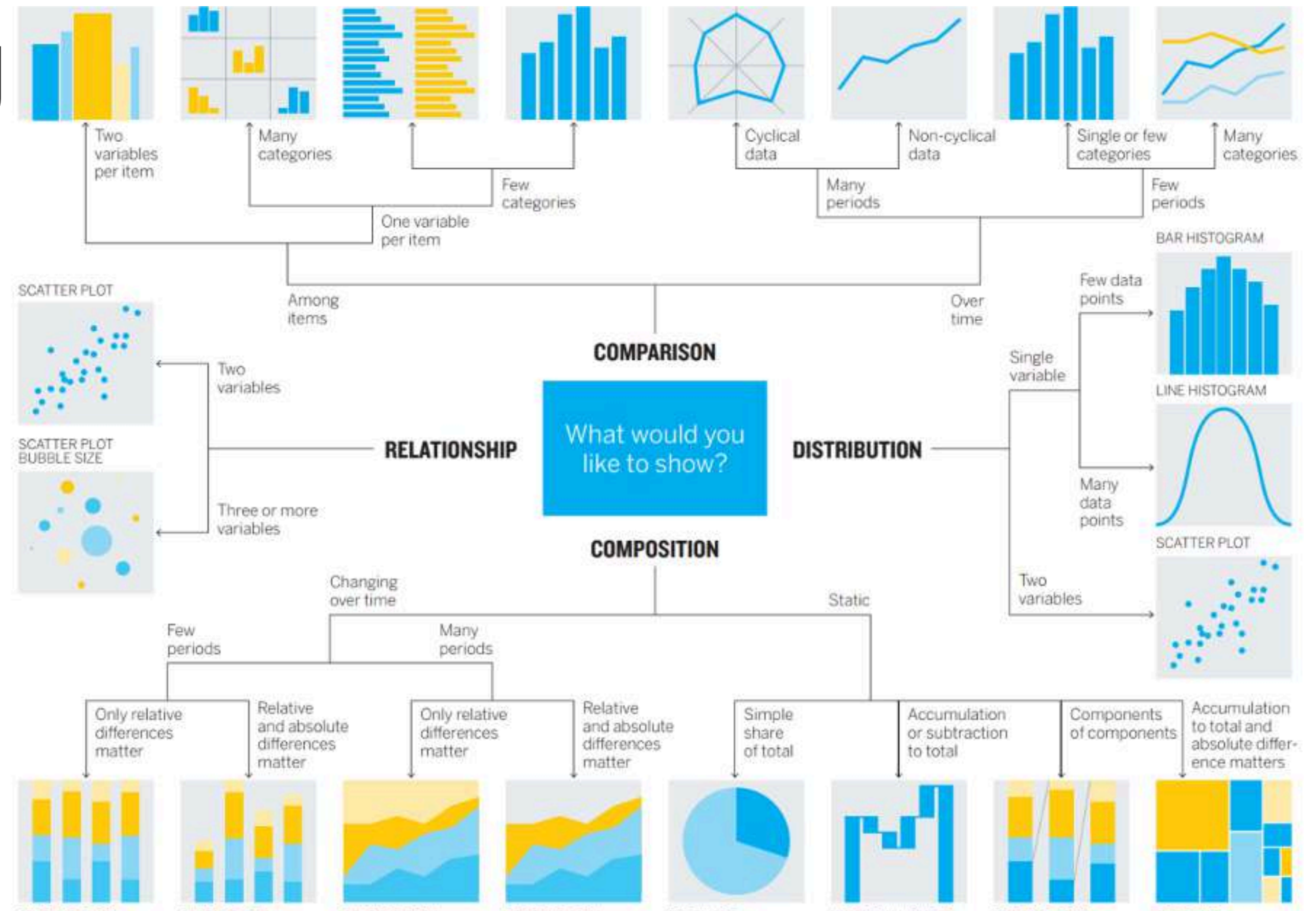
Alerting
color

Be careful

Not to become the next Vincent van Gogh...



Choosing the right chart



A Good Chart

- 1 Title, introduction, & source
- 2 Measurement, units, scales, & legends
- 3 Methods of visual encoding
- 4 Annotations

“Good charts make us smarter.

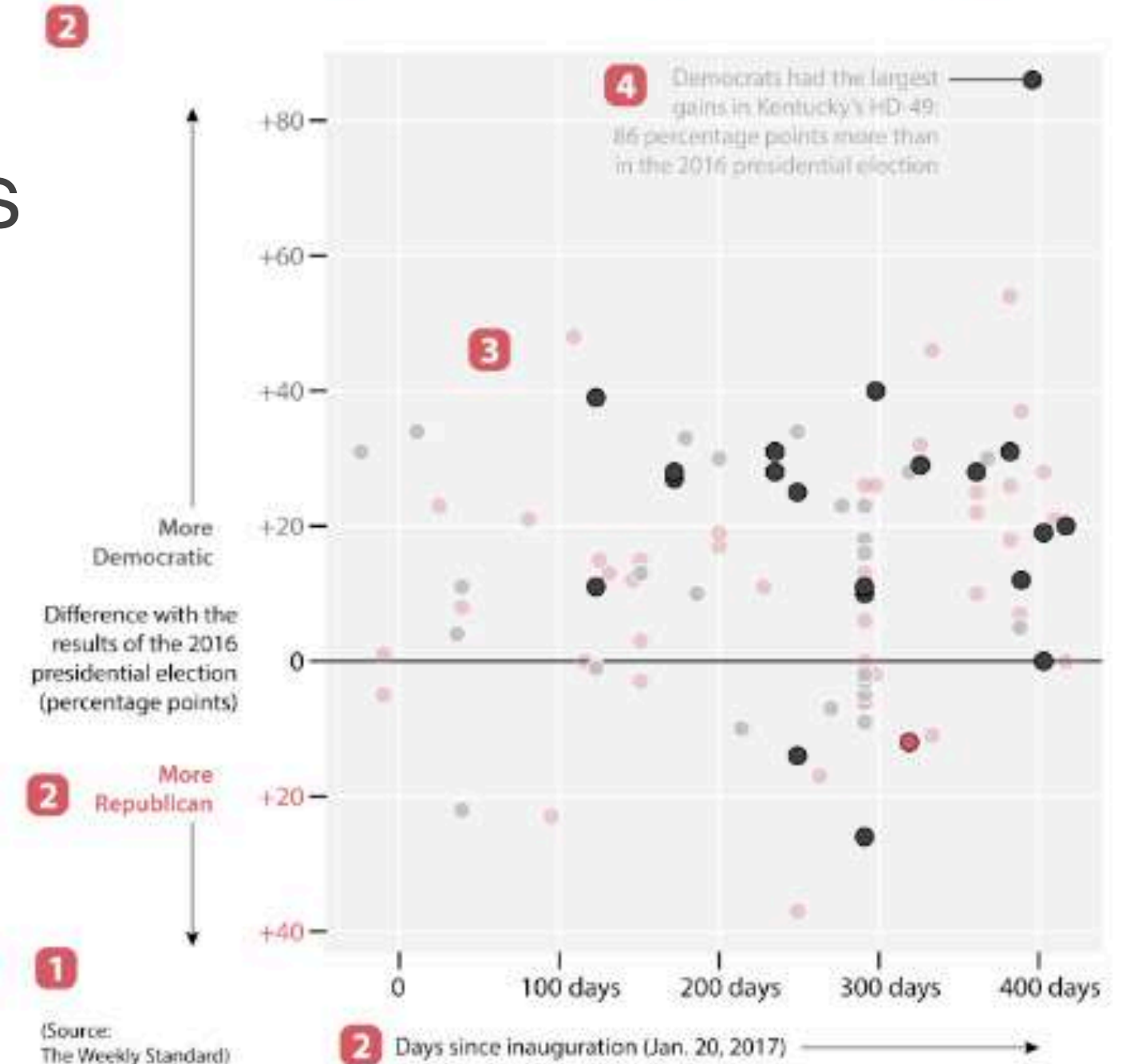
- Cairo, A. (2019)

Democrats are winning special elections ¹

Since Inauguration Day, in January 2017, Democratic candidates have made big gains in special elections. Numerous districts experienced a large increase of democratic votes in comparison to the results of the 2016 presidential election.

Each circle is
a special election

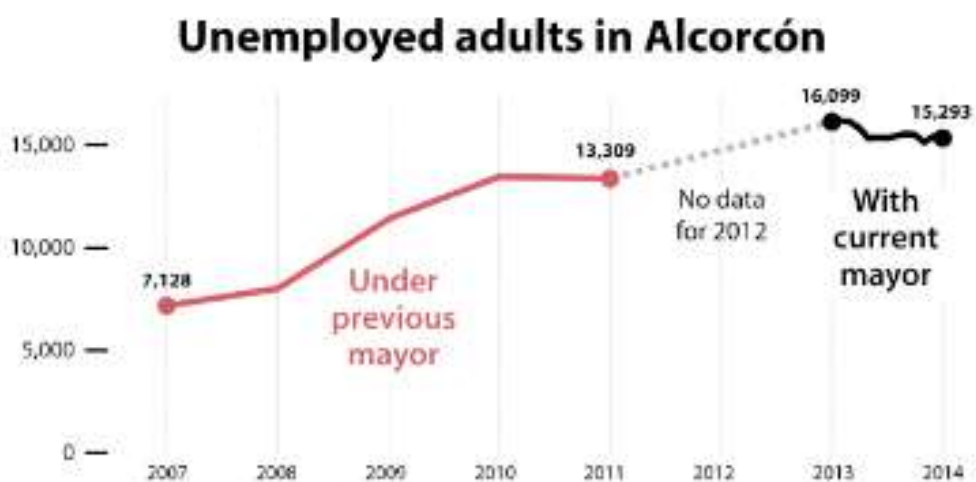
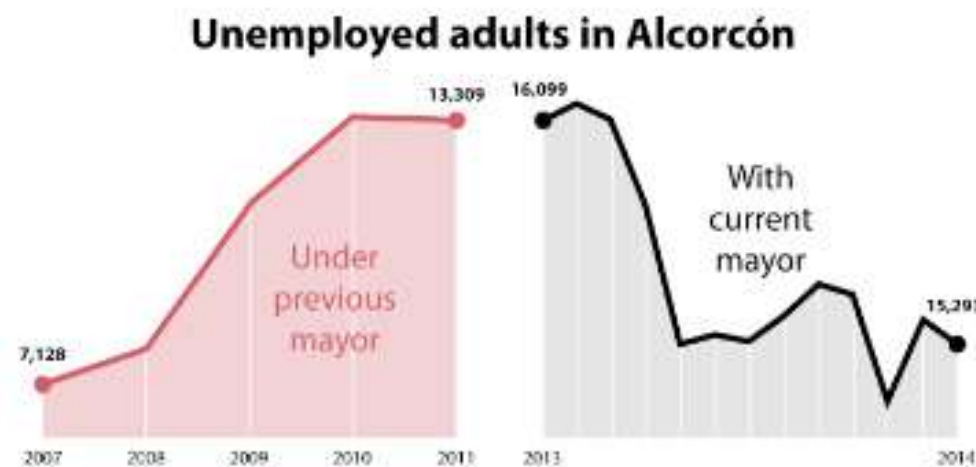
● Democrats flipped the seat ● Democrats held the seat
● Republicans flipped the seat ● Republicans held the seat ²



How Charts Lie

Cairo, A. (2019)

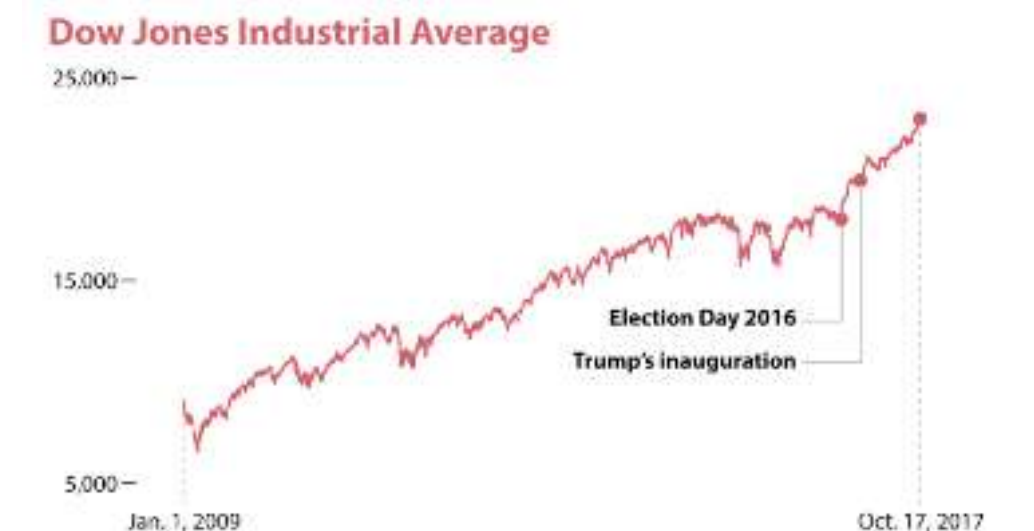
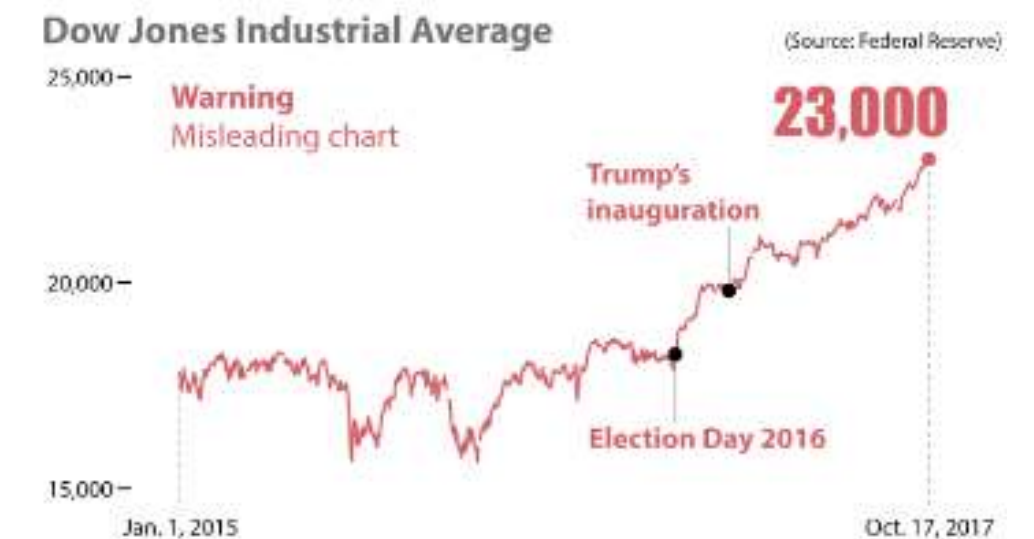
By being poorly designed



By displaying insufficient data



By suggesting misleading patterns



Dashboard

“Is a visual display of data used to monitor conditions and/or facilitate understanding.

- Wexler et al. (2017)
The Big Book of Dashboards

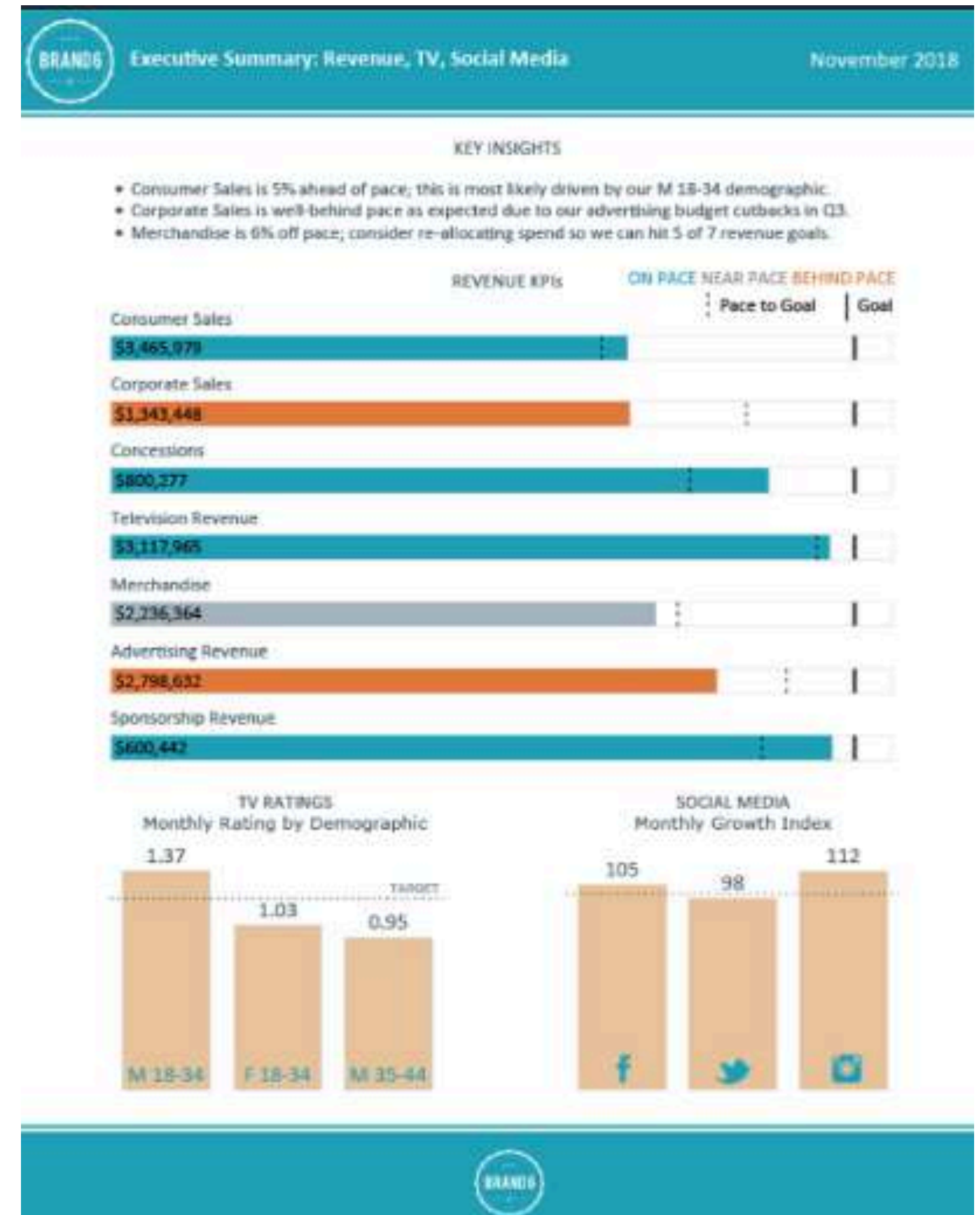
Dashboard types

1. Strategic Dashboard

Strategic dashboards are meant for executives, who want lot of view of their key performance indicators, digestible at a glance as they march from one meeting to the next.

The at-a-glance nature means these dashboards will have a pretty **simple visual design**.

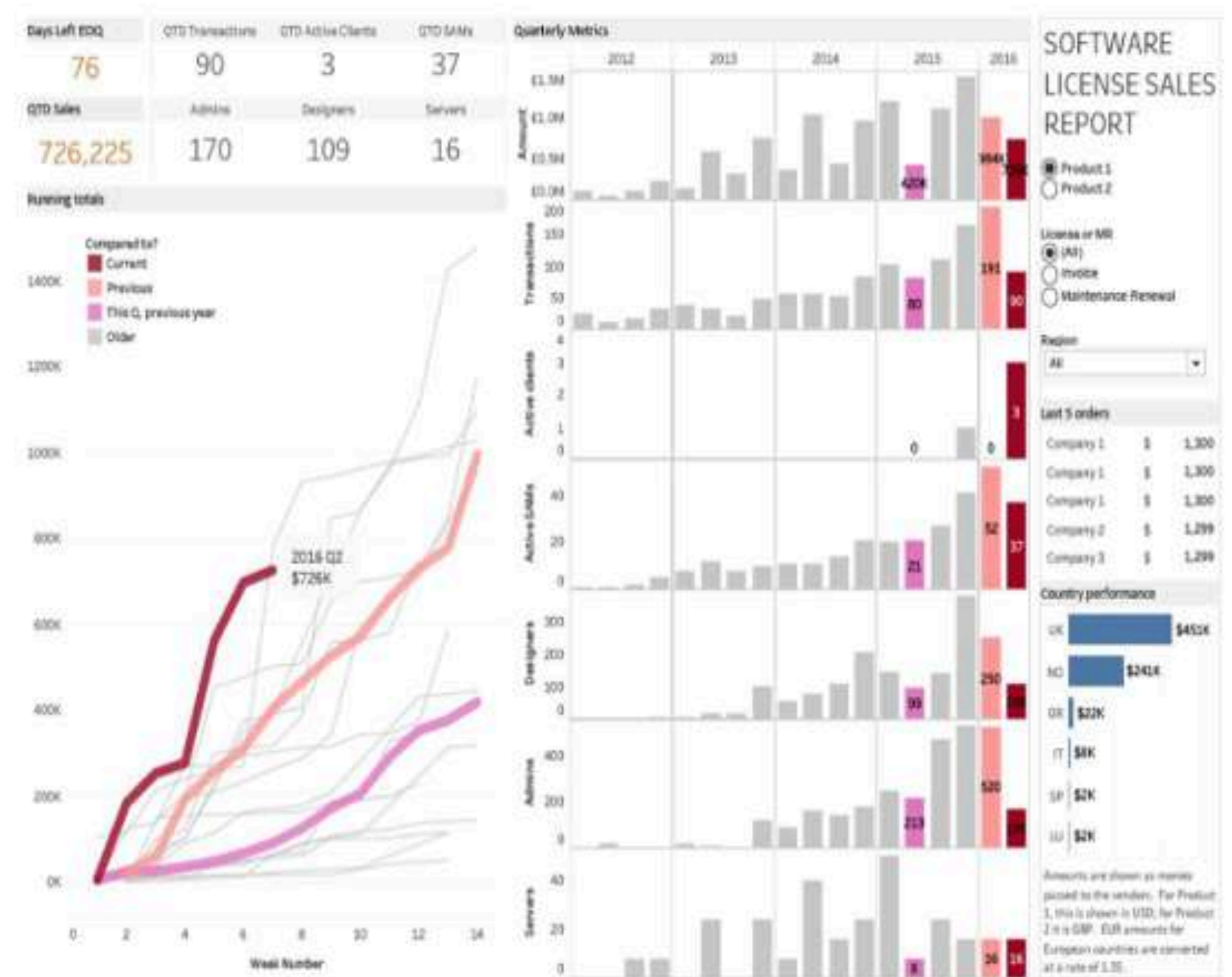
No widgets or dropdowns because an executive doesn't want to have to dig for details.



Dashboard types

2. Analytics Dashboard

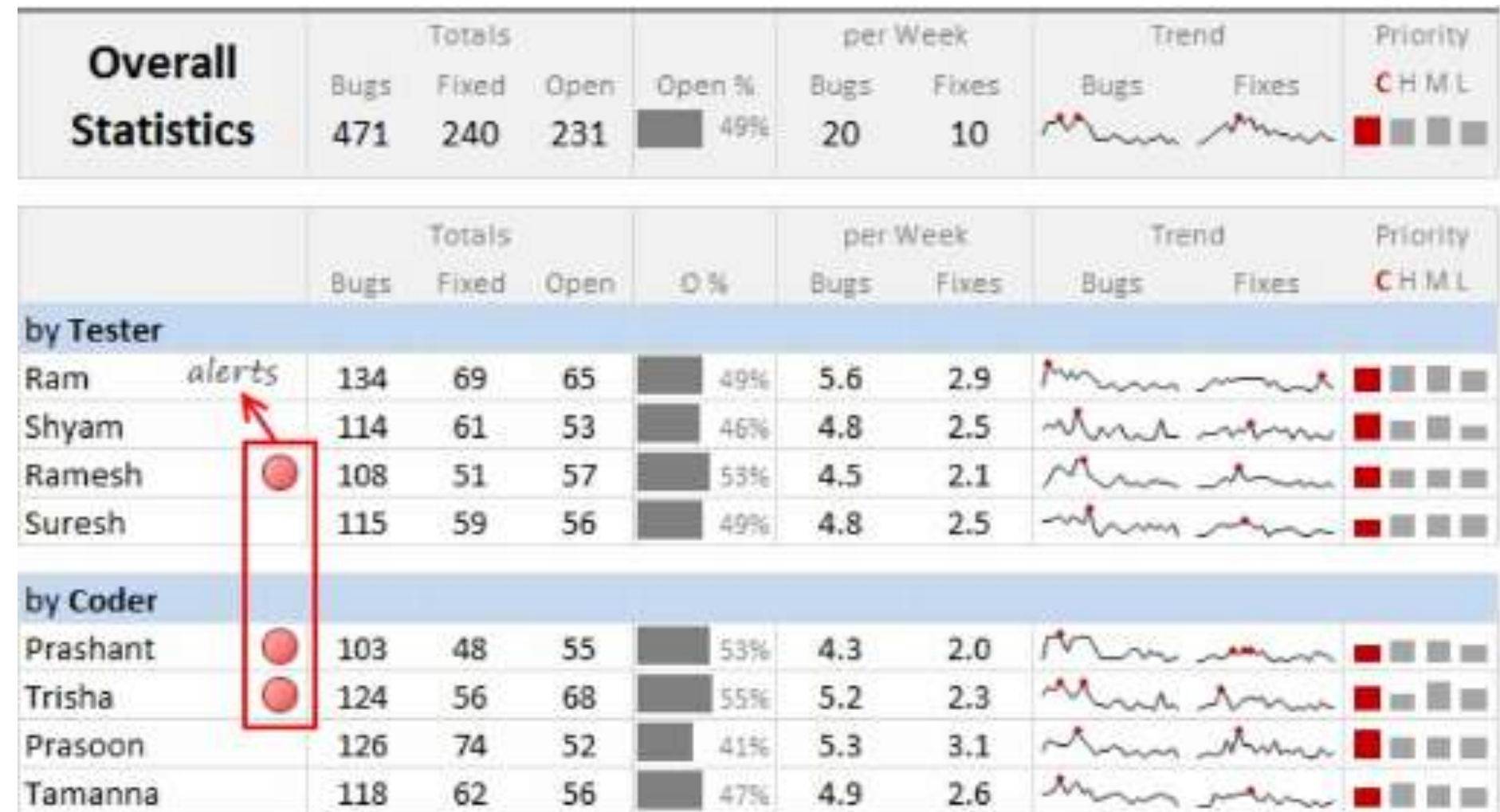
- Have dropdowns and filters to allow for exploration.
- More complex and the front end may have multiple pages or layers so the analyst can drill down and look for patterns and relationships.
- Have all those BANs (Big-A** Numbers) at the top.
- Making comparison, reviewing histories, evaluating performances.



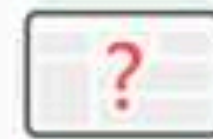
Dashboard types

3. Operational Dashboard

- Have a live connection to the data, meaning the dashboard updates in real time.
- Very simplistic design.
- Monitor function which need constant attention.
- There are some charts but they're mostly gray, meant for context.
- Use highlight color to draws attention where it's needed.



Dashboard Making Guidelines



Focus on specific questions

Avoid creating a report that tries to show everything.



Go from top-left to bottom-right

Place the most important information in the top-left of the report.



Use color to steer attention

Make it quick and convenient for users to get what they need.



Limit ink and information

Avoid overwhelming users with too much to see and think about.



Keep it simple

Choose the right charts for the data, users, and questions.



Make it convenient

Help users spend as little time on the report as possible.

The 3-30-300 rule



3 seconds:

Users should get an overview of the most important questions and areas.



30 seconds:

Users should filter and zoom to identify periods and categories to focus on.



300 seconds:

Users should get details-on-demand to inform their decisions and actions.

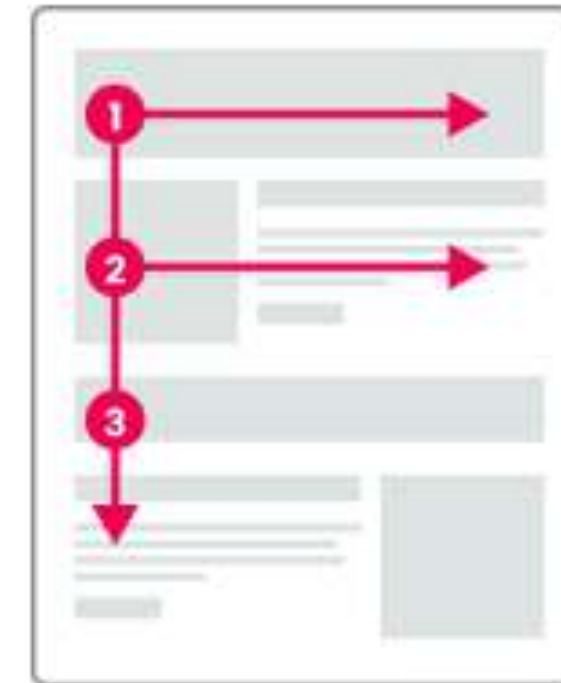
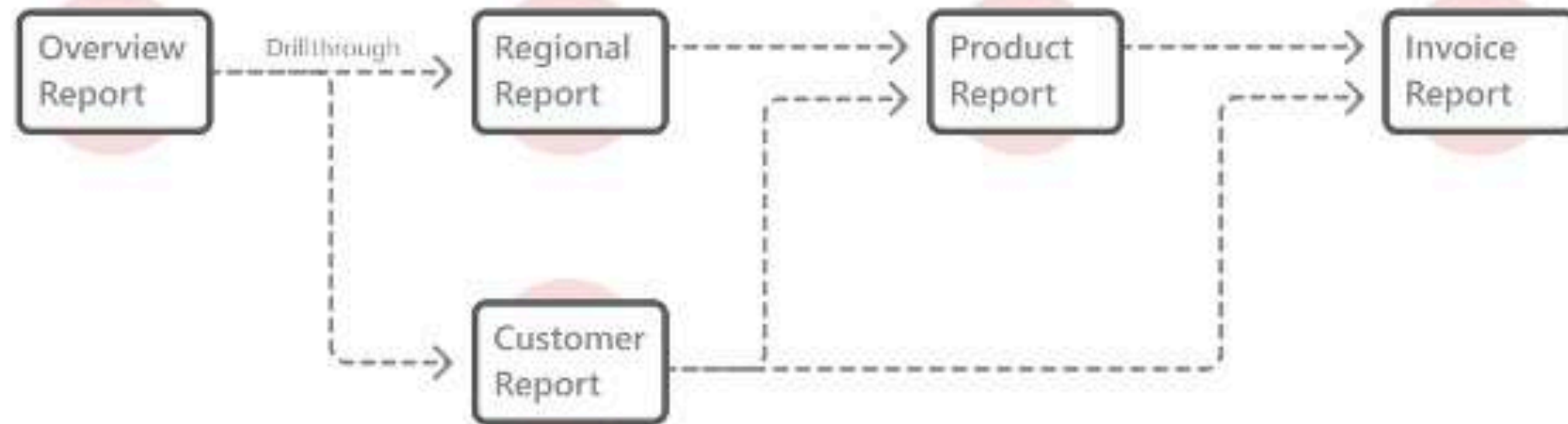
Dashboard Flow

Detail

A flow that depicts how users navigate reports to answer their questions.



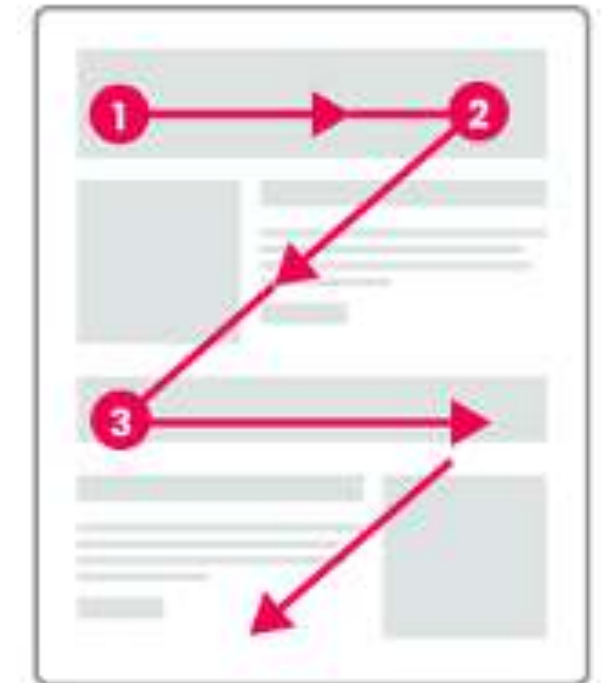
Reports or report pages in a single workspace or app



F-Pattern



✓ Logical flow

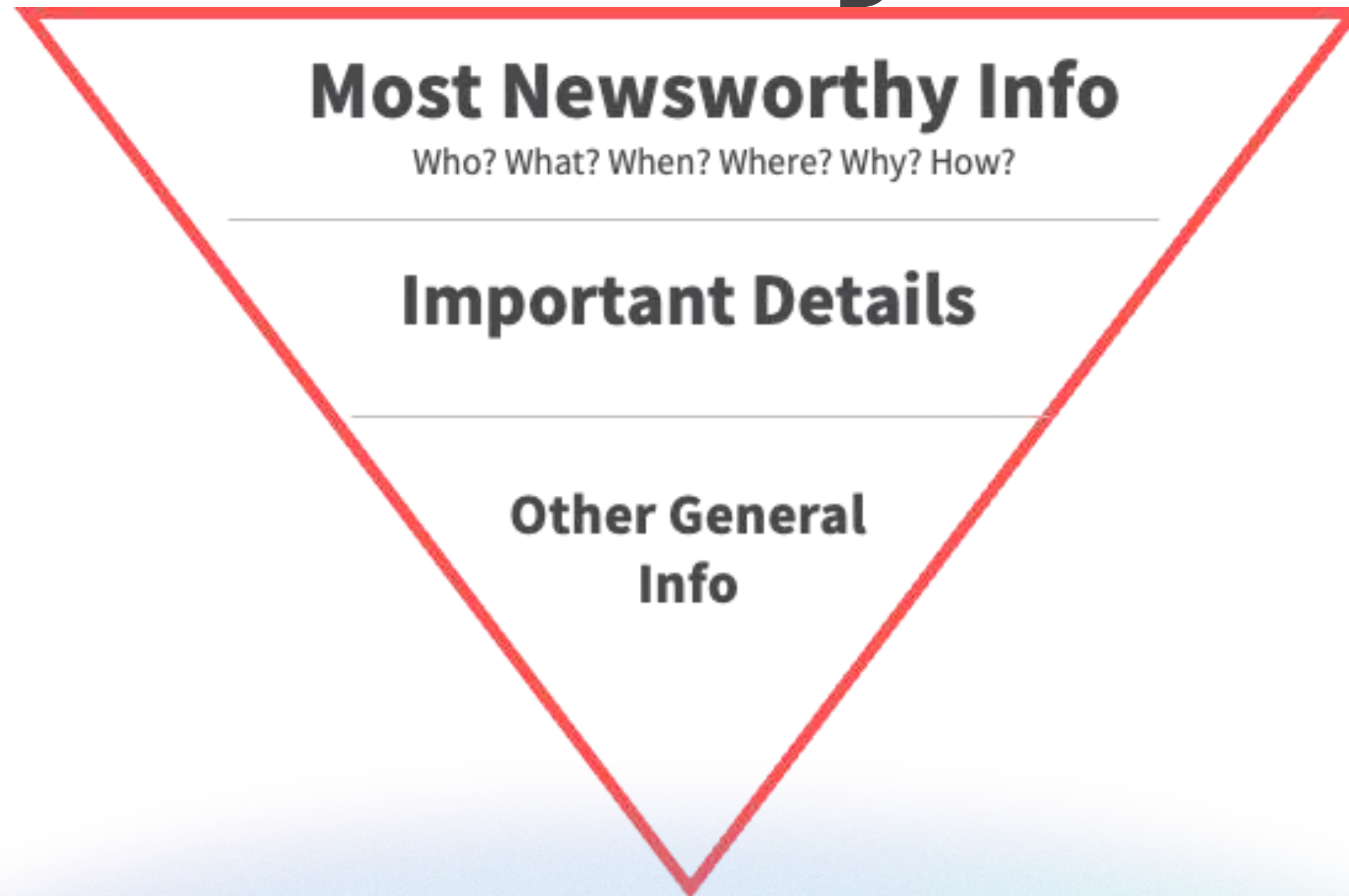


Z-Pattern



✗ Disorderly flow

Inverted Pyramid



Don't be afraid of...



✓ White space and fills



✗ Clutter and gridlines

WHITE SPACE

Dashboarding Tools

Figure 1: Magic Quadrant for Analytics and Business Intelligence Platforms



Pros: free, integrates well with Google services, user-friendly drag & drop, and good for real-time collaboration

Cons: limited visualization options, limited data modeling

Best for: developer and engineering teams in the Google Cloud ecosystem



Pros: broader visualization and data modeling, integrates with Microsoft ecosystem, has advanced AI tools

Cons: steeper learning curve (DAX), expensive licensing

Best for: Microsoft-centric organizations, scalable self-service BI



Pros: industry-leading visualization and strongest visual storytelling tool with great community support

Cons: expensive, self-service data prep

Best for: data analysts and BI teams where visual depth and dashboard quality are the top priority

**be creative
be unique
be you!**

But remember, you have supervisors
who have different tastes
in colors and designs.

“There are no perfect
dashboards.

- Wexler et al. (2017)
The Big Book of Dashboards

When a good viz is not enough...



Data Storytelling

“No one ever made a decision because of a number. They need a story.

- Kahneman, D. (2016)

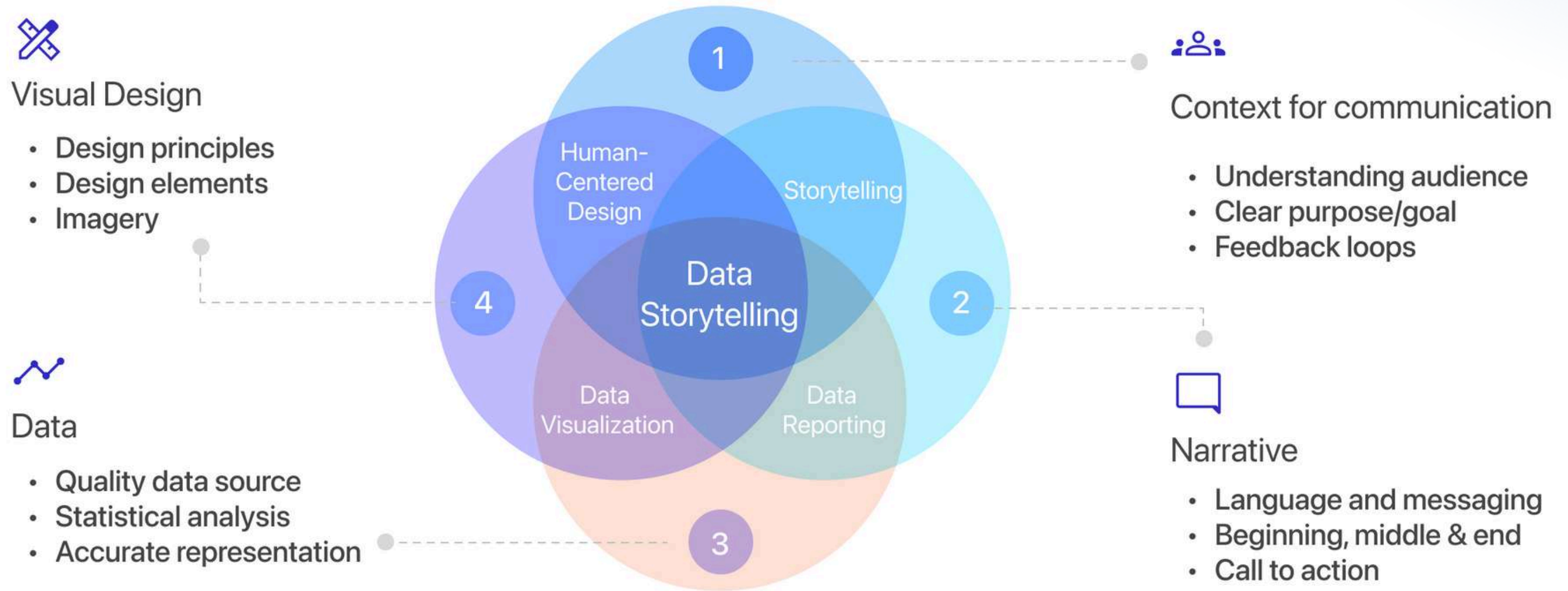
How Two Trailblazing Psychologists Turned the World of Decision Science Upside Down

Why are some people bad at convincing people even with data?



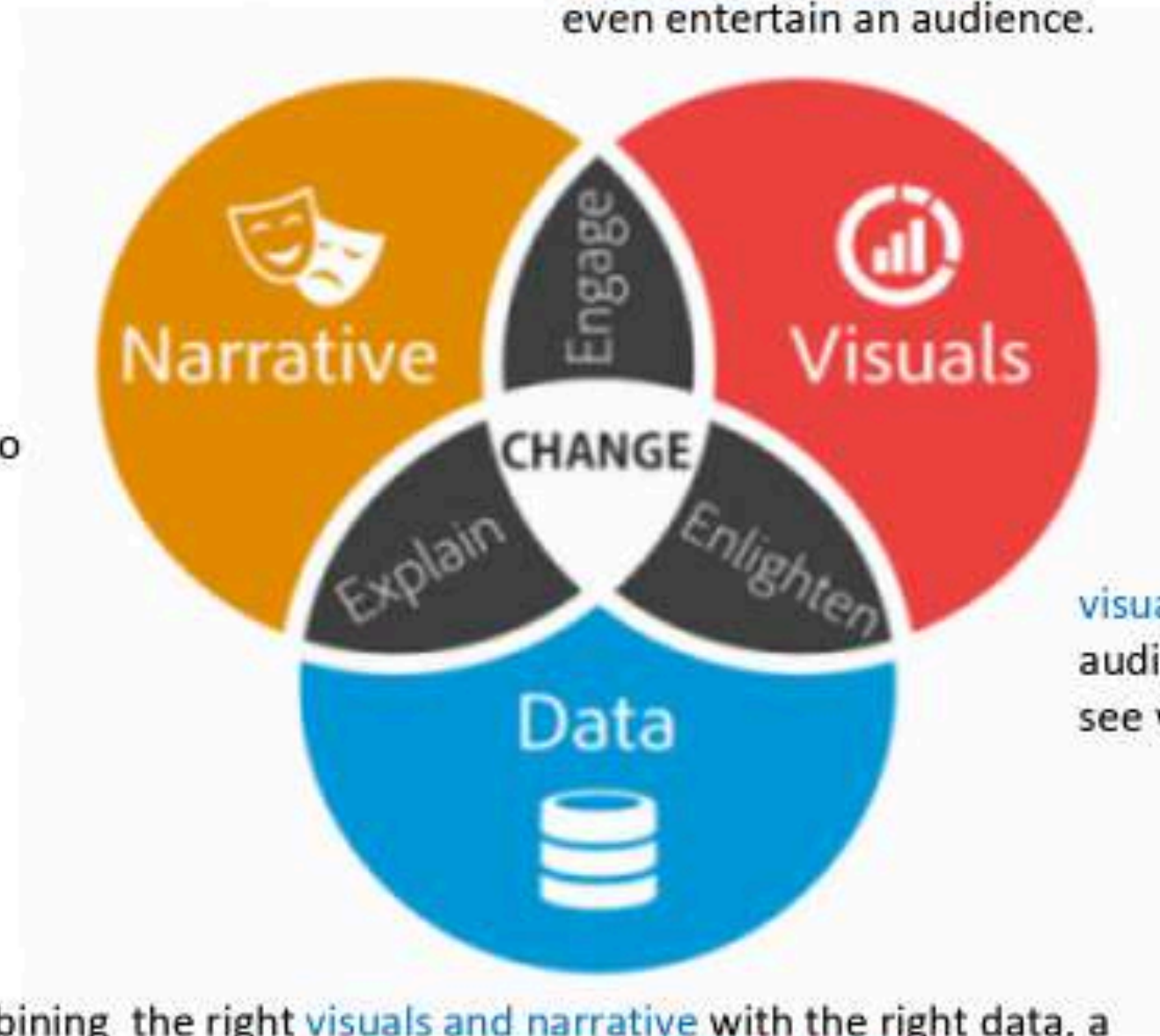
- Analytics people often **aren't that motivated** or successful at communicating with other people.
- They probably **don't get** a lot of instruction to telling data stories in school
- To indulge in storytelling, some analysts may believe, is **an insult**, or at least a relatively less valuable investment of time in comparison, to an analysts' technical capabilities.
- It takes **a lot of analysts' time** to think creatively about how to tell a good story with data.

Data Storytelling



Initiating Changes

narrative + data, it helps to **explain** to your audience what's happening in the data and why a particular insight is important. Ample context and commentary is often needed to fully appreciate an insight. visualizations.



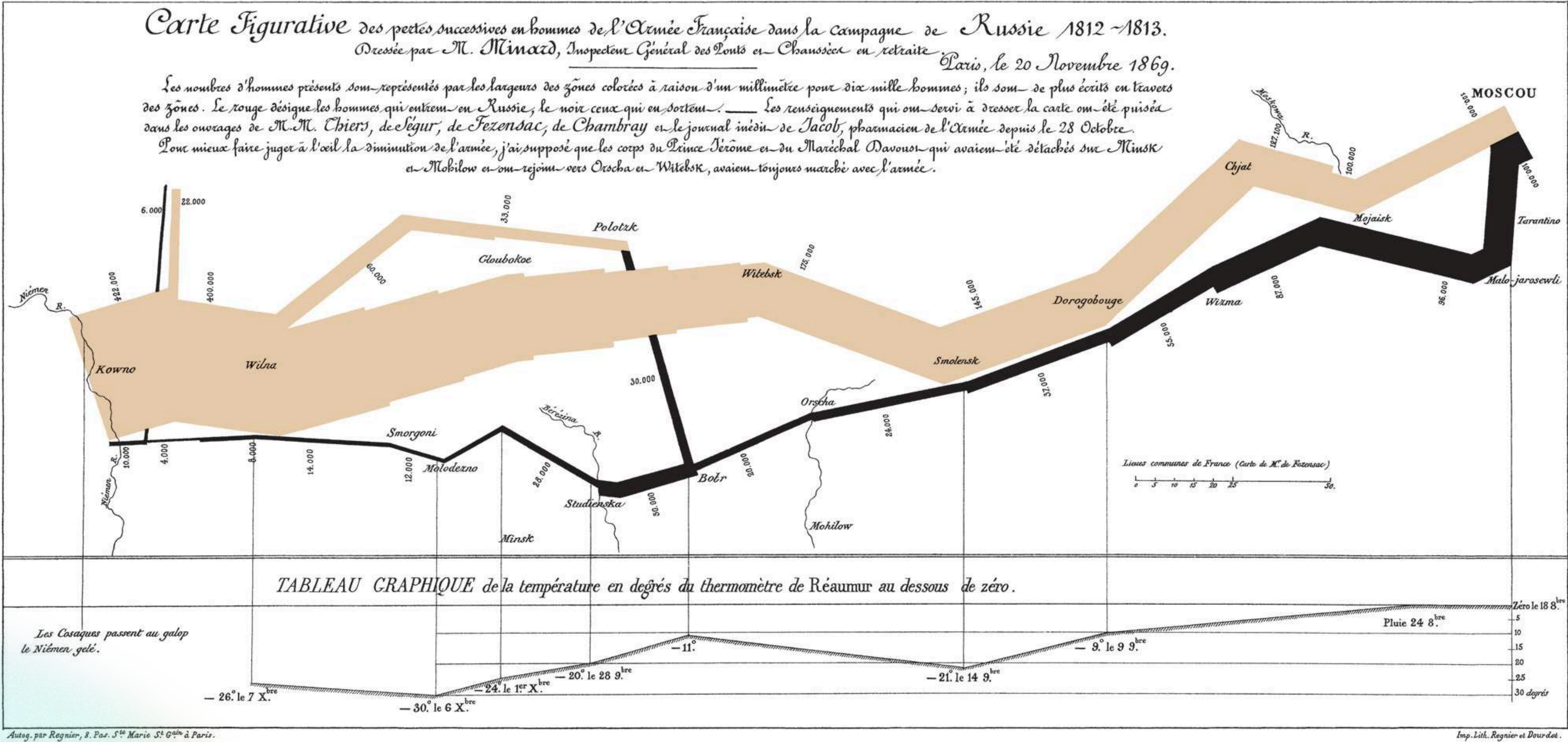
visuals + data, they can **enlighten** the audience to insights that they wouldn't see without charts or graphs.

Combining the right **visuals** and **narrative** with the right data, a data story that can **influence and drive change**.

“People hear statistics, but they feel stories.

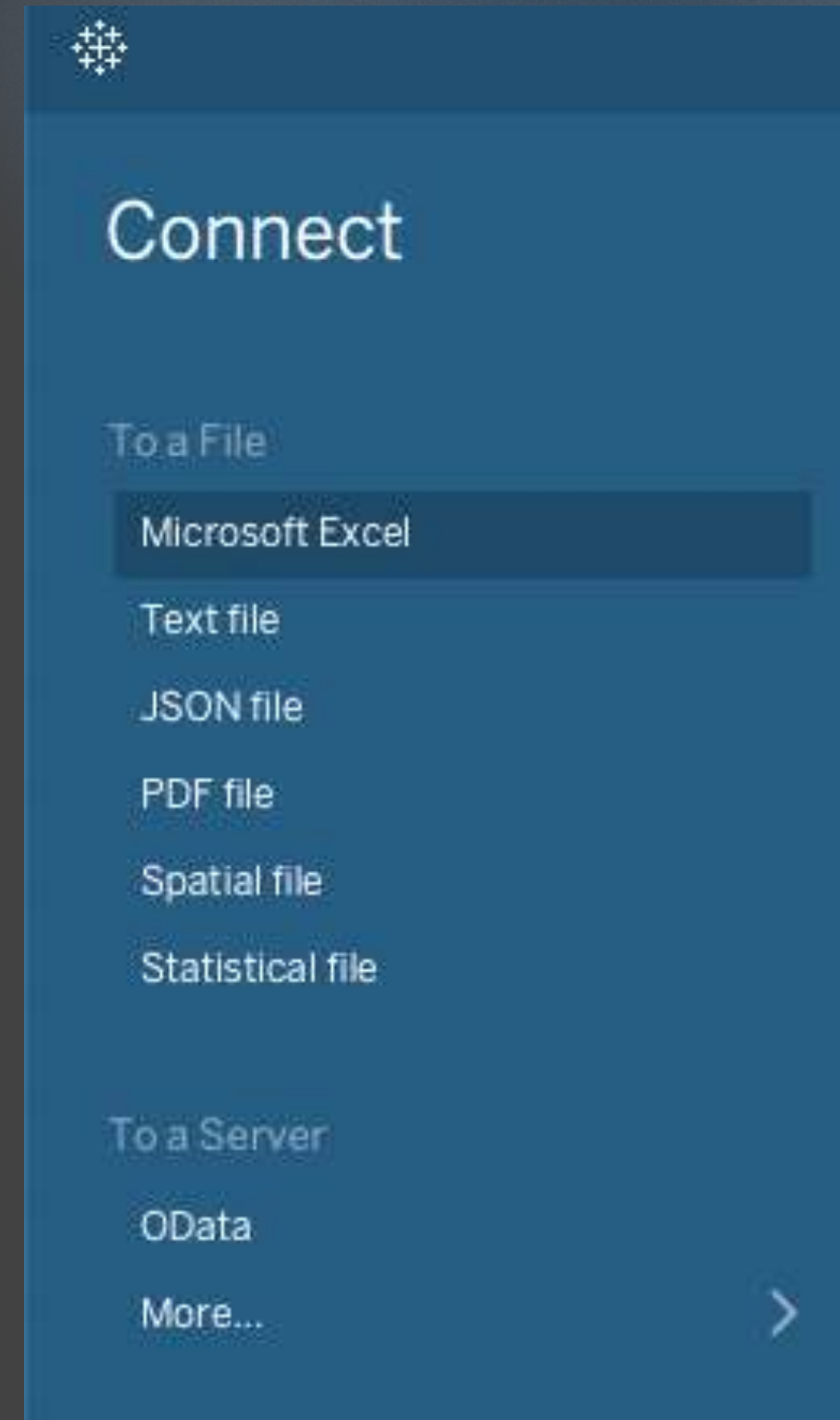
- Dykes, B. (2016)

Charles Minard's 1869 chart of Napoleon's March

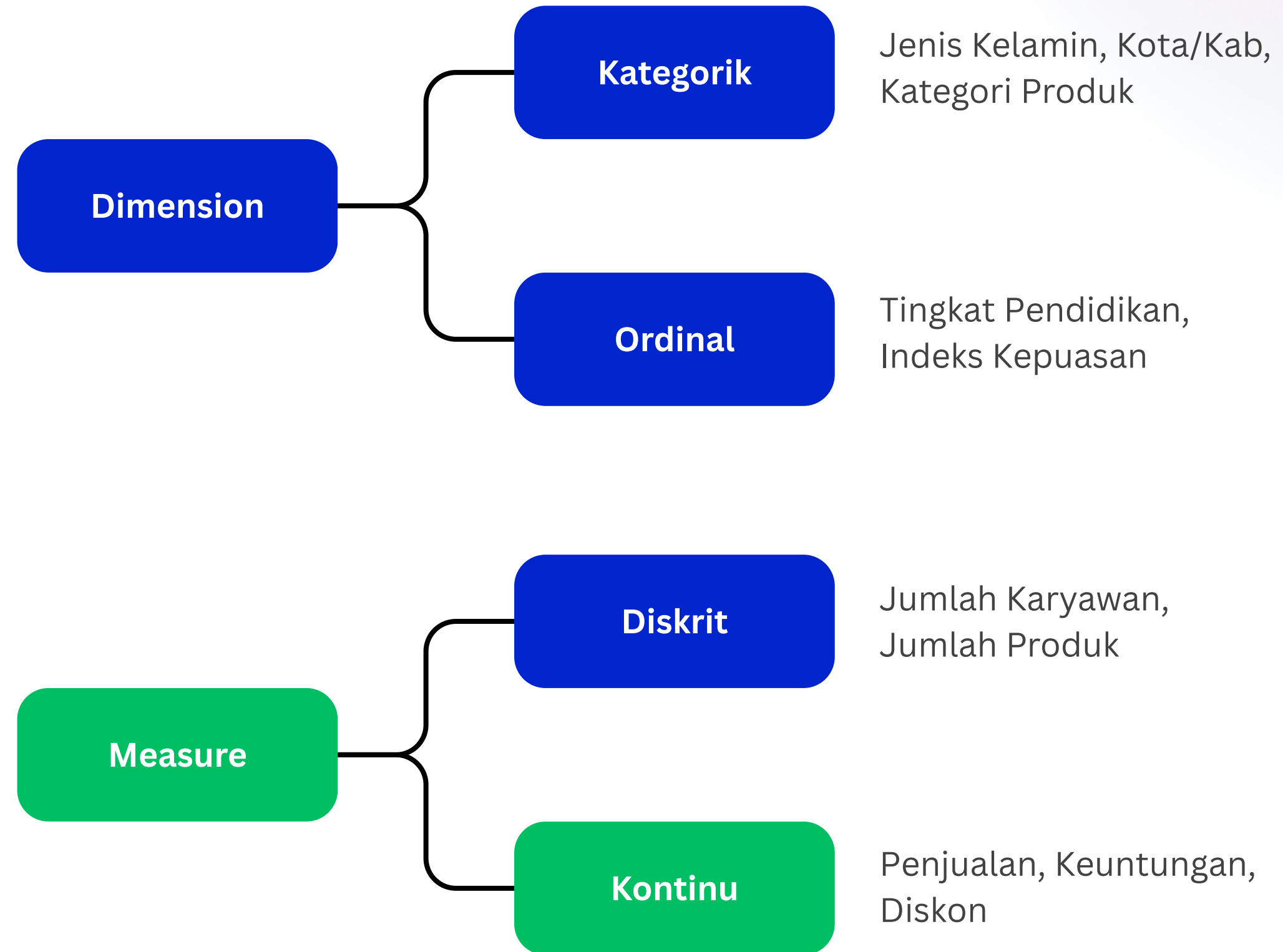


Tufte, E. R. (2001). *The visual display of quantitative information*

Hands-on Tableau Public



Tipe Data





**Thank
You**

for your time
and attention